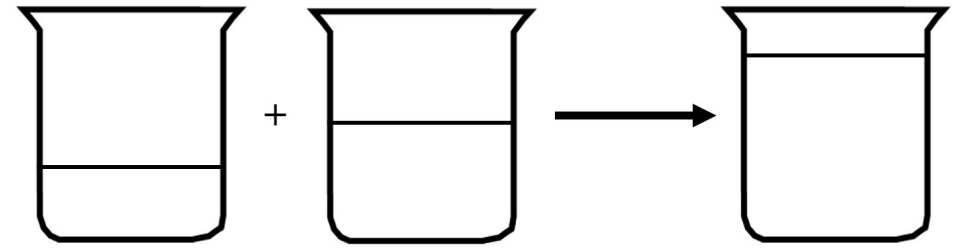
Topics 8.4 – 8.5: MCQ Practice



1. A student combined 300.0 mL of 0.050 *M* $HNO_3(aq)$ with 500.0 mL of 0.032 *M* KOH(*aq*) and mixed the final solution thoroughly. The pH of the resulting solution should be closest to

   (A) 2.90

   (B) 7.00

   (C) 8.30

   (D) 11.10

$$OCl^-(aq) + HF(aq) \rightleftarrows HOCl(aq) + F^-(aq) \qquad K_{eq} > 1$$

2. What are the relative strengths of the acids and bases in the reaction represented by the equation above?

	Acid Strength	Base Strength
(A)	HF < HOCl	F^- < OCl^-
(B)	HF < HOCl	F^- > OCl^-
(C)	HF > HOCl	F^- < OCl^-
(D)	HF > HOCl	F^- > OCl^-

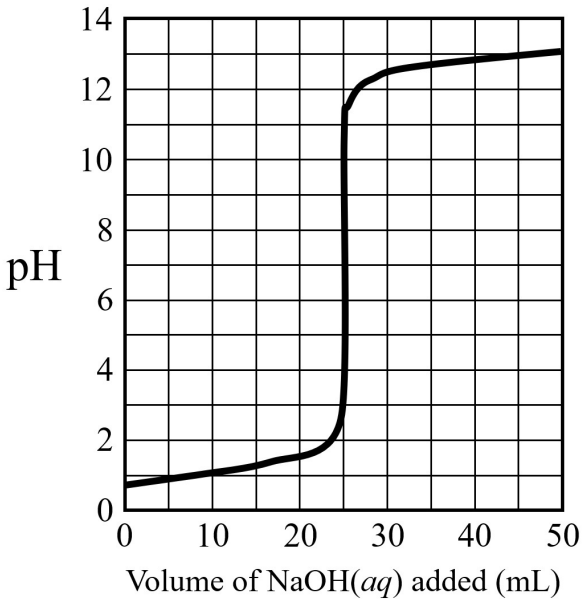
3. A solution of 0.10 *M* HNO_2 ($pK_a = 3.40$) is titrated with 0.20 *M* NaOH(*aq*). Which of the following identifies the pH at the equivalence point of the titration and provides the correct justification?

(A) pH = 7, because the concentration of NaOH is equal to that of HNO_2.

(B) pH = 7, because the NO_2^- ion is classified as a neutral ion.

(C) pH > 7, because the NO_2^- ion behaves as a weak base in the solution.

(D) pH > 7, because the concentration of NaOH is greater than that of HNO_2.



4. A 50.0 mL sample of a 0.20 *M* monoprotic acid solution is titrated with a NaOH(*aq*) solution of unknown concentration. Based on the titration curve above, what is the molar concentration of the NaOH(*aq*) solution?

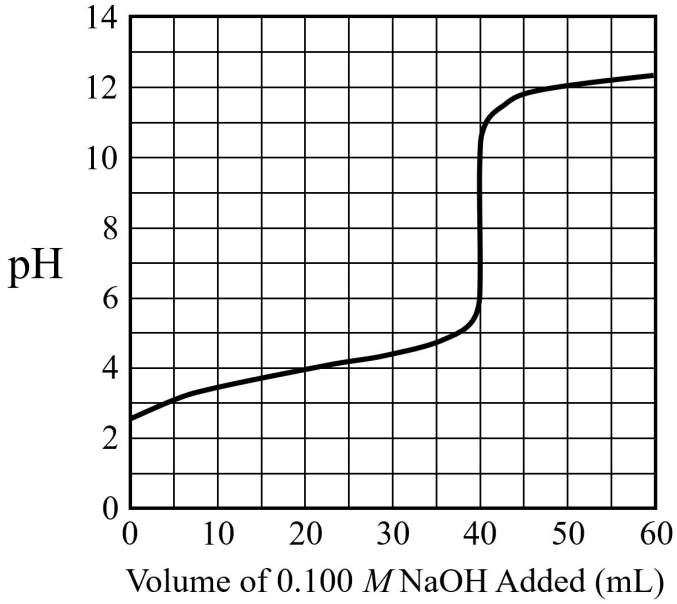
(A) 0.10 *M*

(B) 0.20 *M*

(C) 0.40 *M*

(D) 0.80 *M*

Questions 5 and 6 refer to the following information.



A 50.0 mL sample of a monoprotic acid, HA, of unknown molarity is titrated. The pH of the resulting solution is measured with a pH meter and graphed as a function of the volume of 0.100 *M* NaOH added. The experimental data are shown in the graph above.

5. A student carries out the same titration, but uses an indicator instead of a pH meter. If the indicator changes color slightly past the equivalence point, what will the student obtain for the calculated concentration of the acid?

   (A) Slightly less than 0.0800 *M*

   (B) Slightly more than 0.0800 *M*

   (C) Slightly less than 0.125 *M*

   (D) Slightly more than 0.125 *M*

6. Based on the titration curve above, which of the following could be the identity of the weak acid?

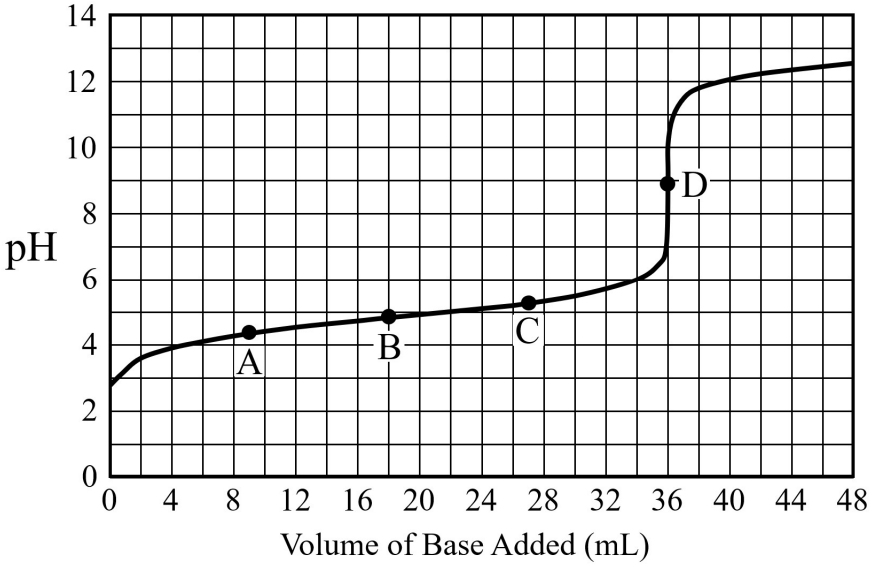
   (A) HOI with a K_a value of 2.3×10^{-11}

   (B) HOCl with a K_a value of 2.9×10^{-8}

   (C) $HC_3H_5O_3$ with a K_a value of 1.4×10^{-4}

   (D) $HClO_2$ with a K_a value of 1.1×10^{-2}

Questions 7 and 8 refer to the following information.

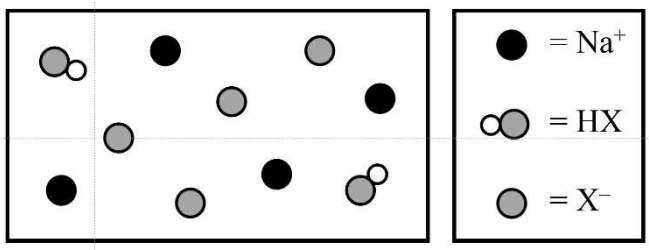


A student performs an acid-base titration of a monoprotic acid HX and plots the experimental results in the graph above.

7. Which of the following provides the correct information about this titration experiment?

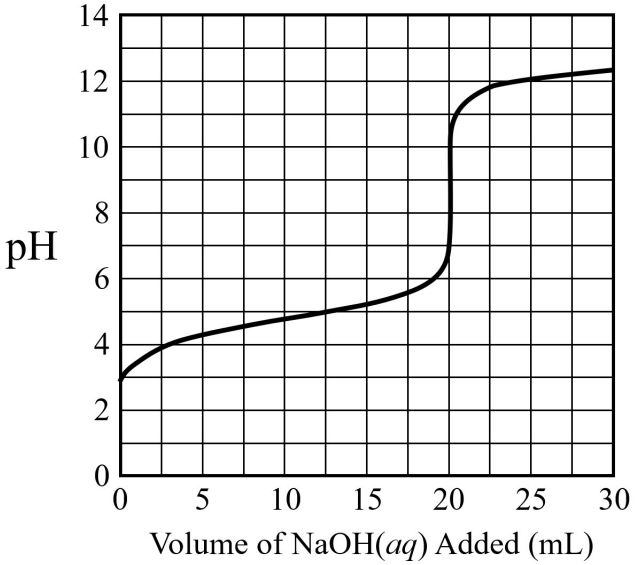
	Description of the Reactants	pH at the Equivalence Point of the Titration
(A)	A strong acid was titrated with a strong base.	pH = 7
(B)	A strong acid was titrated with a strong base.	pH > 7
(C)	A weak acid was titrated with a strong base.	pH = 7
(D)	A weak acid was titrated with a strong base.	pH > 7

8. The particle diagram shown at right represents the relative amounts of major species in a sample of the solution in the flask at one point during the titration. (Note that water molecules are omitted.)



Which of the points labeled on the titration curve is most likely to represent the point in the titration where the reaction mixture would be represented by this particle diagram?

(A) Point A (B) Point B (C) Point C (D) Point D



9. Data collected during the titration of a 20.0 mL sample of a 0.10 *M* solution of a monoprotic acid with a solution of NaOH of unknown concentration are plotted in the graph above. Based on the data, which of the following are the approximate pK_a of the acid and the molar concentration of the NaOH?

	pK_a	[NaOH]
(A)	4.7	0.050 *M*
(B)	4.7	0.10 *M*
(C)	9.3	0.050 *M*
(D)	9.3	0.10 *M*

Solution	Acid	K_a
1	CH_3CO_2H	1.75×10^{-5}
2	CF_3CO_2H	1.0×10^{0}

10. Acid-dissociation constants of two acids are listed in the table above. A 20. mL sample of a 0.10 *M* solution of each acid is titrated to the equivalence point with 20. mL of 0.10 *M* NaOH. Which of the following is a true statement about the pH of the solutions at the equivalence point?

(A) Solution 1 has a higher pH at the equivalence point because CH_3CO_2H is the stronger acid.

(B) Solution 1 has a higher pH at the equivalence point because CH_3CO_2H has the stronger conjugate base.

(C) Solution 1 has a lower pH at the equivalence point because CH_3CO_2H is the stronger acid.

(D) Solution 1 has a lower pH at the equivalence point because CH_3CO_2H has the stronger conjugate base.

Experimental data for two different monoprotic acids are listed in the table below.

Acid	Concentration of Acid Solution (M)	Volume of Acid Solution (mL)	pH of Acid Solution
$HClO_4(aq)$	0.10	50.0	1.00
$HClO(aq)$	0.10	50.0	4.27

11. Two different titration experiments are performed, in which a 50.0 mL sample of each acid, at a concentration of 0.10 M, is titrated with 0.10 M $NaOH(aq)$. Which of the following statements correctly compares the volume of $NaOH(aq)$ required to reach the equivalence point in each titration?

    (A) $HClO_4$ will require a greater volume of $NaOH(aq)$ to reach the equivalence point.

    (B) $HClO$ will require a greater volume of $NaOH(aq)$ to reach the equivalence point.

    (C) Each acid will require the same volume of $NaOH(aq)$ to reach the equivalence point.

    (D) The volume of $NaOH(aq)$ required to reach the equivalence point cannot be determined without knowing the values of K_a for $HClO_4(aq)$ and $HClO(aq)$.

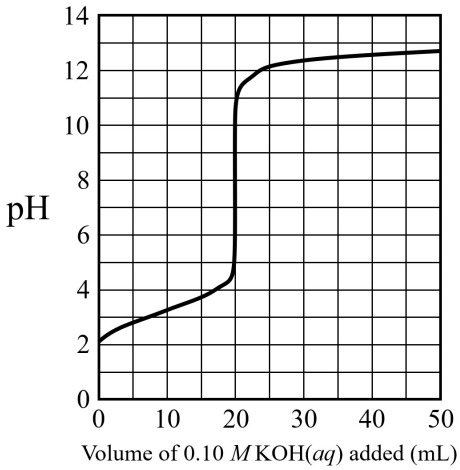
12. A solution containing HBr and the weak acid $HC_2H_3O_2$ has a pH of 3.0. Enough $NaOH(aq)$ is added to the solution to increase the pH to 11.0. The amount of which of the following species increases as the $NaOH(aq)$ is added?

    (A) $Br^-(aq)$

    (B) $H^+(aq)$

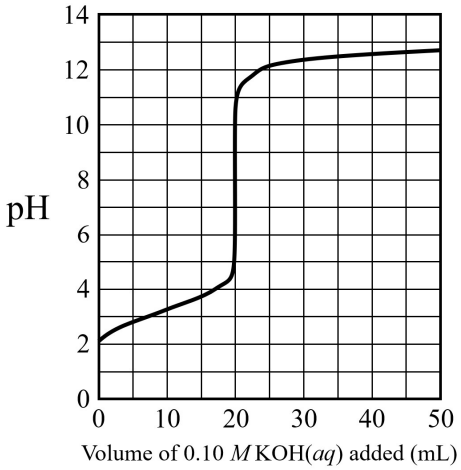
    (C) $C_2H_3O_2^-(aq)$

    (D) $HC_2H_3O_2(aq)$

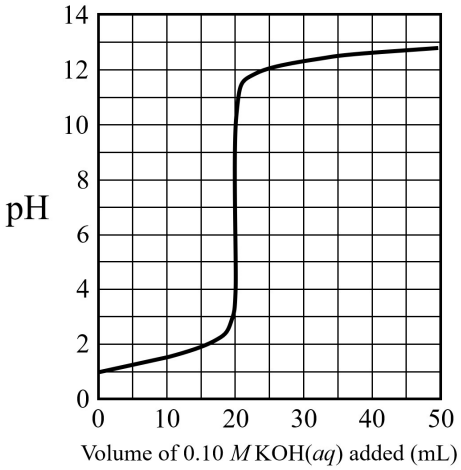


13. A 20.0 mL sample of 0.10 *M* HF(*aq*) is titrated with 0.10 *M* KOH(*aq*), and the data are shown in the graph above. Which of the following graphs shows the change in pH when a 20.0 mL sample of 0.10 *M* HNO_3(*aq*) is titrated with 0.10 *M* KOH(*aq*)?

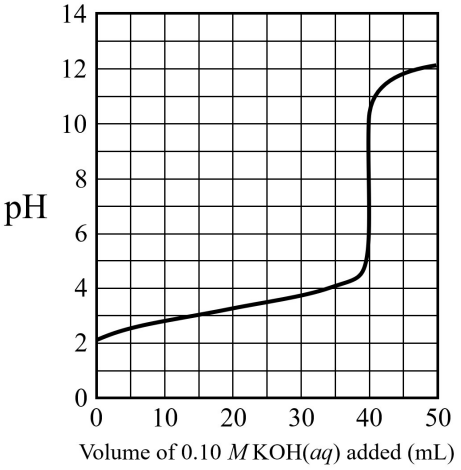
(A)



(C)



(B)



(D)

